

User Manual

NS_ND24Series Thermocouple Temperature Transmitter

Temperature transmitter with Thermocouple, NS_ND24 Series

NS24T,ND24T

NS244T,ND244T

NS2408T,ND2408T



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1、Product Introduction

NS24The series of thermocouple temperature transmitters are independently developed, tested and produced by Lingshuo Technology.

The series of products have been used in various industrial sites and experimental research sites, with high stability, high safety, high

Cost-effectiveness and other characteristics.

This series of products include2road,4Road and8The road can be chosen, Communication interface isRS485,supportK/S/R/T/N/E/J

The default parameters of the product after production testing areKType thermocouple, temperature range

The range supports full range temperature measurement. The specific models are described in Table 1. If it does not meet the actual use requirements,

You can contact us for customized products.

Table 1

Number of channels/ Function	Two channels	Four channels	Eight channels	Remark
Basic	NS24T	NS244T	NS2408T	defaultKtype
Display	ND24T	ND244T	ND2408T	defaultKtype

1.1.Product Features

- 2~8Optional collection channels
- The update rate of the collected temperature value is1Hz/aisle
- standardRS485 MODBUS-RTUprotocol
- Support local temperature detection (cold end compensation)
- Power supply, communication and collection three-terminal isolation (1500VDC)
- Full temperature measurement range:-200~1370°C (KType), other types of temperature ranges are supported See4.8
- high resolution:0.1°C
- Low measurement error:1°C
- Built-in watchdog to prevent system crashes, never downtime
- passCEQuality and safety certification

1.2.Features

- High speed, high precision and high stability
- 32BitARMHigh-speed processor main control chip
- 16High precision dedicated temperatureADSampling chip
- Equipped with channel indicator light and high-definition digital display screen (optional)
- High conversion efficiency, using10-30VDCUltra-wide power input voltage
- Strong protection, the power supply has anti-reverse connection protection, over-current and over-voltage protection
- Strong anti-interference, power supply, communication and collection three-terminal isolation, can be passed3500VGroup pulse interference test
- Easy to install, standard35mmUniversal rail installation
- Industrial-grade products that can meet the needs of various fields

1.3.Product Parameters

parameter	Minimum	Standard value	Maximum	unit	Remark
Operating voltage	+ 10		+ 30	V	DC
Power consumption		0.2	0.3	W	+ 24VDC power supply
Operating temperature	- 20	+ 25	60	°C	
Operating humidity	0		95	RH	No condensation
AB line voltage	+ 3	+ 3.5	+ 4	V	Not connected to the bus
Update rate			1	s	Per channel
Isolation voltage			1500	VDC	
Temperature error			1	°C	
Measurement error			1	°C	
Resolution		0.1		°C	
Communication Protocol					Modbus RTU
Communications equipment		32		individual	
Communication station number	1		240		Power-off save
Baud rate	1200	9600	115200	bps	Power-off save
Communication distance			1200	rice	
Functional instructions					Read single or multiple registers (Function code 03H)
					Writing a single register (Function code 06H)
					Writing multiple registers (Function code 10H)
Channel 1~8 temperature value (°C) Read register address					0064H~006BH
Parameter reset					Software reset (restart first then reset)
Indicator Lights					Normal operation (green light flashes once every 1s) Read and write data (green light flashes quickly) Return data (red light flashes quickly)
Protection function					Overcurrent/overvoltage/reverse connection
Installation					Standard 35mm universal rail

Notice:

Power access prohibited RS485 The bus will burn out RS485 Communication interface, causing the device to be unable to communicate. Such problems are not free of charge!

1.4.Product dimensions and wiring

1.4.1. NS24T/ND24TDimensions and wiring diagram

Length, width and height:88.5mm(L)*34.2mm(W)*32.3mm(H)

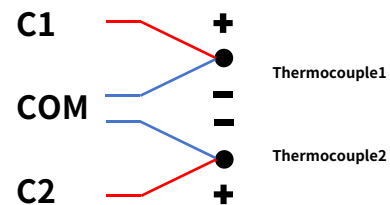


VCC: Power +

GND:power supply-

B:RS485 B

A:RS485 A



1.4.2. NS244T/ND244T Dimensions and wiring diagram

Length, width and height: 82.0mm(L)*54.0mm(W)*32.0mm(H)

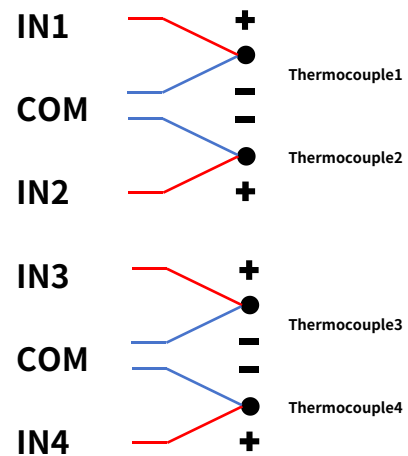


VCC: Power +

GND: power supply-

B: RS485 B

A: RS485 A



1.4.3. NS2408T/ND2408T Dimensions and wiring diagram

Length, width and height:99.0mm(L)*60.0mm(W)*32.0mm(H)



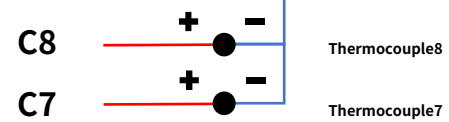
VCC: Power +

GND: power supply-

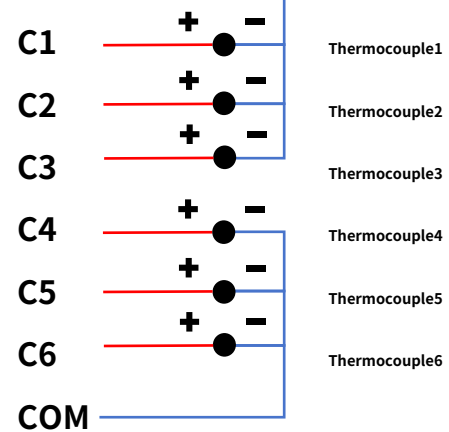
B: RS485 B

A: RS485 A

COM



COM



1.4.4.High-definition digital display function (optional)

The product supports optional channel indicator lights and

Measure the temperature value.

Channel indicator lightCH1,CH2:

When the device is powered on normally, it will

Flashing (interval1s) indicates the current display channel

Digital tube display:

When the device is powered on normally, it will

Flashing (interval1s) indicates the current measured temperature



Power-on display description:

Each time the device is powered on normally, the interface will display the serial port setting parameters.

1, Display the station number of the device and display the time2s.

As shown below, the left picture shows "1" , i.e. station number1.

2, display the baud rate, stop bit, check bit of the device, and display time4s.

As shown below, the right picture shows "300" , that is, the baud rate9600, Stop bit1There is no check digit.

(For specific digital tube display font size definition, refer to2.1Register Definition)



2, Communication Protocol

2.1 Register Definition

Serial port factory default: 9600 8 N 1, device station number: 1; (The following numbers are all decimal)

Register Address	R/W	Remarks (writable range/unit)
1	R/W	Station number (1-240)
2	R/W	Baud rate (0-7)
4	R/W	Stop bit (0-1)
5	R/W	Check digit (0-2)
10	R/W	Thermocouple (0-2: K/S/R) (4-7: T/N/E/J)
37	R/W	Restart the device (write 1 to restart)
99	R	Machine temperature (°C)
100~107	R	Channel 1~8 temperature value (°C)
400~407	R/W	Channel 1~8 offset value (°C)

Note: (Please restart the device after modifying the parameters)

1. Baud rate 0-7 corresponds to 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200
2. Stop bits 0-1 correspond to one stop bit and two stop bits respectively.
3. Check bits 0-2 correspond to no check, odd check, and even check.
4. The data type of temperature value is a 16-bit signed number, and the actual value is one tenth of the register value.
5. The data type of the offset value is an 8-bit signed number, and the actual value is one tenth of the register value.

2.2 Register instructions

2.2.1 Read single or multiple registers (function code 03H)

from 17 Number read from the machine 2 The collected temperature data U1, U2, U1 The address is 0064H (Decimal is 100), U2 The address is 0065H (Decimal is 101).

Addr	Fun	Data start addr hi	Data start addr lo	Data #of regs hi	Data #of regs lo	CRC16 hi	CRC16 lo
11H	03H	00H	64H	00H	02H	XXH	XXH

Each register in the returned information corresponds to 2 bytes, the first byte is the high-order data, and the second byte is the low-order data.

Addr	Fun	Byte count	Data1 hi	Data1 Lo	Data 2 hi	Data2 lo	CRC16 hi	CRC16 lo
11H	03H	04H	01H	02H	FFH	6AH	XXH	XXH

Return temperature data U1 for 0102H, converted to decimal is 258, divided by 10 Get the measured temperature value 25.8°C. Returns temperature data U2 for FF6AH, yes -150 The complement of 10 Get the measured temperature value -15.0°C. (First FF6AH Convert to decimal 65386, then subtract 65536 That is, negative temperature data is obtained -150, and then divided by 10 Get the temperature value -15.0°C)

2.2.2 Write a single register (function code 06H)

Revise 17 Slave serial port baud rate register (02H) has a value of 3 (9600)

Addr	Fun	Data start reg hi	Data start reg lo	Value hi	Value lo	CRC hi	CRC lo
11H	06H	00H	02H	00H	03H	XXH	XXH

2.2.3 Write multiple registers (function code 10H)

Modify at the same time 17 Slave Channel 1, aisle 2 The offset value (0190H, 0191H) has a value of 10 (+1°C)

Addr	Fun	Data start reg hi	Data start reg lo	Data #of reg hi	Data #of reg lo	Byte count	Value hi	Value lo	Value hi	Value lo	CRC hi	CRC lo
11H	10H	01H	90H	00H	02H	04H	00H	10H	00H	10H	XXH	XXH

If you still have questions, please search MODBUS-RTU Protocol learning, or refer to the instructions sent during the debugging software testing provided by our company.

3, debug software

Note: Parameter modification must be completed within two minutes after the module is powered on. It is recommended to restart the device before modification and then configure.

HP2412调试软件
Linso TECH.
专业生产各类变送器控制器

连接设置
COM: COM3, 停止位: 1, 波特率: 9600, 校验位: N, 数据位: 8. 按钮: 刷新COM, 连接.

通讯参数配置
站号: 1, 设置, 停止位: 1, 设置, 波特率: 9600, 设置, 校验位: No, 设置. 按钮: 调试信息, 清空, 复位参数, 重启设备, 全部读取.

数据处理
采集通道数量: 1, 自动读取, 采集时间: 间隔: 1 s, 已保存数据数量: 0, 选择保存路径, 开始保存, 开始保存时间, 曲线监控.

常用功能
设备地址(站号): 1, 本机温度: 0.0 °C, 读取, 全部读取. 温度值1-12: 0.0 °C, 读取, 偏移量1-12: 0.0 °C, 读取, 配置. 传感器类型: PT100, 读取, 配置, 热电偶: K, 读取, 配置, NTC阻值: 10 K, 读取, 配置, NTC B值: 3950, 读取, 配置.

3.1 Device Connection

COM Configuration: Select COMSlogan, baud rate, stop bit, check bit and other parameters, click **connect**.

(The factory default baud rate is 9600, data bits 8, stop bit 1, check digit N, station number 1)

COM配置
COM: COM1, 停止位: 1, 波特率: 9600, 校验位: N, 数据位: 8. 按钮: 刷新COM, 断开.

3.2 Modify communication parameters

When the device communication is normal, fill in the target value for the parameter to be modified and click Set. (If you forget the communication parameters, disconnect other devices on the bus, ensure that the wiring of the device to be reset is normal, and click Reset Parameters)

通讯参数配置
站号: 1, 设置, 停止位: 1, 设置, 波特率: 9600, 设置, 校验位: No, 设置. 按钮: 调试信息, 清空, 复位参数, 重启设备, 全部读取.
△ 修改(复位)参数需在上电两分钟内完成, 超时请在修改前重启设备

3.3 Equipment debugging

Fill in the device address (station number) to be operated, and read the temperature or configure the device parameters through this page.

常用功能

设备地址(站号) 本机温度

温度读取

温度值1	0.0 °C	读取	偏移量1	0.0 °C	读取	配置
温度值2	0.0 °C	读取	偏移量2	0.0 °C	读取	配置
温度值3	0.0 °C	读取	偏移量3	0.0 °C	读取	配置
温度值4	0.0 °C	读取	偏移量4	0.0 °C	读取	配置
温度值5	0.0 °C	读取	偏移量5	0.0 °C	读取	配置
温度值6	0.0 °C	读取	偏移量6	0.0 °C	读取	配置
温度值7	0.0 °C	读取	偏移量7	0.0 °C	读取	配置
温度值8	0.0 °C	读取	偏移量8	0.0 °C	读取	配置
温度值9	0.0 °C	读取	偏移量9	0.0 °C	读取	配置
温度值10	0.0 °C	读取	偏移量10	0.0 °C	读取	配置
温度值11	0.0 °C	读取	偏移量11	0.0 °C	读取	配置
温度值12	0.0 °C	读取	偏移量12	0.0 °C	读取	配置

Offset: The temperature read is equal to the measured temperature plus the offset value. For example, the device measures a temperature of 50°C, the user sets the offset value to +10(+1°C), then the temperature data read is 510, converted to temperature is 51.0°C.

3.4 Modify sensor type

Thermal resistors and thermocouples cannot be modified and can only be selected when purchasing. You can check which category the device input type belongs to by looking at the label on the product surface.

传感器类型读取与设置

热电阻 热电偶

NTC 阻值 NTC B值

热电偶类型选择: K, S, R, B, T, N, E, J

If you need to change the thermocouple type, select the type under the clicked model.

3.5 Automatically read and save data

When the device communication is normal, it can be set to automatically read and save temperature data. (To use the data saving function, you need to install WPS or Office wait EXCEL Table viewing function software)

Number of acquisition channels: Customizable number of channels for acquisition 1-240.

Collection time: Customizable collection intervals 1-60s.

Save data: Click "Auto Read" and select the path to save the data file; click "Start Saving" and the debugging software will automatically

Automatically save the read message information and display the amount of saved data; finally, "Stop saving" will be generated in the corresponding path

become **Excel Format** of temperature information file, open it to view the saved data.

记录时间	通道1	通道2	通道3	通道4	通道5	通道6	通道7	通道8
2024-10-17 11:24:46	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:47	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:48	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:49	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:50	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:51	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:52	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:53	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:54	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:55	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:56	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:57	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:58	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:24:59	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:25:00	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:25:01	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:25:02	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:25:03	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:25:04	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1
2024-10-17 11:25:05	850.4	850.4	850.4	850.4	850.1	850.1	850.1	850.1

4, Notes

4.1 Communication distance:

Standard RS485 Theoretically, the communication distance can reach 1200 meters, actual distance and on-site interference, the number of devices on the bus

If the communication distance is long, it is recommended to use a shielded RVV line, or use a repeater to amplify the signal.

4.2 Communication protocol: Standard MODBUS-RTU Protocol. Refer to the register table.

4.3 Sensor distance:

Three-wire system PT100 The resistance of the sensor extension cable can be calculated to subtract the error caused by the cable resistance.

use 25AWG, then each 100m The resistance of the extension cord is about 80Ω, the maximum line resistance allowed by our equipment is 400Ω, then

The sensor extension cable can theoretically be connected 500 meters. (Make sure the extension cords are made of the same wire material and have the same length. Otherwise,

error.)

4.4 Two-wire system PT100 Can the sensor replace the three-wire system? PT100? How to connect?

The answer is yes, but it will introduce the error of the extension line resistance. Therefore, if the detection distance is short, the error is within the acceptable range.

If the input is within the range, it can be replaced, otherwise it is not recommended. When wiring, just connect the negative input terminal to COM. Short-circuit the sensor

Just connect the two wires to the positive input terminal and the negative input terminal respectively. PT100 Can also replace two lines PT100, just

To short two wires of the same color, PT100 Negative, connected to the negative input terminal or COM end.

4.5 The data read is 0xFFFF (65535): If the data read is 65535, then it means that the channel sensor

Disconnect. Does not return when the program is set to work normally 65535 The value of the measured temperature should indeed be returned 65535, equipment

Will add by default 1 Then return. That is, if the measured temperature is -0.1°C, then the temperature value returned is 0°C.

4.6 How to reset serial port parameters: If the serial port parameters or device address are modified and communication fails, use the debugging software

COM Reset the parameters in the configuration. (Make sure the device is normal and 485 The communication wiring is correct.)

4.7 Do I need to recalibrate after changing the sensor type? No calibration is required to achieve the nominal accuracy.

4.8 Comparison table of various types of thermocouples with full range temperature

K Type	- 270°C~1370°C
S-type	50°C~1700°C
R Type	50°C~1700°C
Type B	0°C~1800°C
T-type	- 270°C~400°C
N-type	- 200°C~1300°C
Type E	- 270°C~1000°C
J Type	- 200°C~1200°C